MIL-R-51197B(EA)
15 September 1976
SUPERSEDING
MIL-R-51197A(MU)
6 August 1969

MILITARY SPECIFICATION

REAGENTS, SOLVENTS, AND TEST PAPERS FOR SAMPLING AND

ANALYZING KIT, CBR AGENTS

This specification is approved for use by Edgewood Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers chemical components used in the analyzing of chemical agents.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378	-	Plastic Film (Polyethylene, Thin Gage).
O-C-265	-	Chemicals, Analytical; General Specification for.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-676	-	Boxes, Set-Up.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-P-291	-	Paperboard, Wrapping, Cushioning.
PPP-T-60	-	Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.

MILITARY

MIL-P-116	-	Preservative, Methods of.
MIL-B-117	-	Bags, Sleeves and Tubing - Interior Packaging.
MIL-I-285	-	Impregnite, Unstabilized (CC2) and Stabilized (XXCC3).
MIL-R-5001	-	Rubber Cellular Sheet, Molded and Hand Built Shapes, Latex Foam.

FSC 6665

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MIL-B-11845 - Bromthymol Blue Solution.
MIL-R-20092 - Rubber Sheets and Molded Shapes Cellular, Synthetic, Exploded Cell (Foam Latex).
MIL-P-26514 - Polyurethane Foam, Rigid or Elastic, for Packaging.
MIL-P-50014 - Packet, Sodium Pyrophosphate Peroxide.
MIL-D-50028 - DB3.
MIL-T-51023 - Tablets, Compound 34.
MIL-C-51130 - Chemicals, Reagent Grade; General Specification For.
MIL-S-51189 - Substrate Solution and Dispenser.

STANDARDS

FEDERAL

FED-STD-356 - Commercial Packaging of Supplies and Equipment.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

US ARMY ARMAMENT COMMAND

EDGEWOOD ARSENAL

B5-77-209 - Packet, White, Filled.
B5-77-442 - Paper, Dragendorff.
B5-77-447 - Tube, Alcohol.
B5-77-448 - Vial, Glass.
B5-77-485 - Vial, Packet Filling.
C5-77-441 - Packet, Combination Filter.
C5-77-444 - Tube, Benzene.
C5-77-456 - Bottle, Squeeze.
DLB 136-41-780-1 - Dilution Apparatus, Toxic Gas, Q5.

PUBLICATIONS

EDGEWOOD ARSENAL PURCHASE DESCRIPTIONS

197-54-505 - Di(p-Biphenyl)-Thiocarbazone (DBT).

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on the applicable component drawings (see table II) shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.1.1 Reagents. All chemicals used to prepare reagents shall conform to the requirements of O-C-265. Where O-C-265 is not applicable, reagent grade chemicals shall be used except where otherwise specified. The supplier shall certify that the reagents are no older than 6 months at time of delivery, except for sodium pyrophosphate peroxide and sodium perborate which shall be no older that 120 days.

(a) Modified Dragendorff reagent. The reagent is made by mixing 15.2 grams (g) of metaphosphoric acid into 100 milliliters (ml) of distilled water, then adding and stirring until dissolved 6.7 g of cadmium iodide and 8.6 g of calcium chloride, and finally mixing 20 g of sorbitol thoroughly in the solution.

(b) 4,4'-Bis(diethylamino) benzophenone. The 4,4'-bis(diethylamino) benzophenone shall have a melting point of 91° ± 2° C and comply with one of the following Munsell color ranges:

(a) 2.5Y, 9 value, minimum (min), 6 chroma min;
(b) 5.0Y, 9 value min, 4 chroma min; and
(c) 7.5Y, 9 value min, 2 chroma min.

The 4,4'-bis (diethylamino) benzophenone shall be considered satisfactory if a drop of reagent 20 placed on Whatman No. 1 filter paper in a test tube and heated in a water bath for 2 minutes, gives no red or red-brown color.

(c) Sodium perborate. The sodium perborate tetrahydrate ($NaBO_3 \cdot 4H_2O$) shall conform to the chemical requirements of table I.

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TABLE I. Chemical requirements of sodium perborate

Property	Percent by weight	
	Minimum	Maximum
Available oxygen	10.0	---
Chloride	---	0.001
Phosphate	---	0.001
Sulfate	---	0.003
Heavy metals (as lead)	---	0.002
Iron (Fe)	---	0.001
Silicate	---	0.002

The sodium perborate shall be thoroughly mixed with sodium chloride in the proportions of 1 milligram (mg) of sodium perborate to 15 to 20 mg of the sodium chloride and this quantity placed in a packet conforming to Drawing B5-77-209. The particle size of each material shall be the same.

(d) Ortho-dianisidine dihydrochloride (ODN.HCl). The ODN.HCl (see 6.3) shall have a relative purity of 92 to 100 percent when tested as specified in 4.3.4.1(a)(1). An ODN.HCl sodium pyrophosphate peroxide solution (8.00 ± 0.50 mg ODN.HCl and 2.000 ± 0.200 g sodium pyrophosphate peroxide in 100 ml water) shall have a minimum transmittance of 95 percent at a wavelength of 510 millimicrons when tested as specified in 4.3.4.1(a)(1).

3.1.1.2 Solvents. The benzene used to fill the benzene tubes (Dwg C5-77-444) shall consist of a mixture of benzene conforming to the requirements of O-C-265 and 10 ± 0.1 percent of ligroin, (Density 0.68 - 0.70/20° C, Boiling Point 66 - 75° C). The mixture shall have a freezing point of less than minus 2° C.

3.1.2 Components. The components shall be as shown in table II.

TABLE II. Components (Reagent, Solvent and Test Paper)

Components	Number per kit	Drawing
Reagent container sets	2	Table III
Modified Dragendorff papers	75	B5-77-442
Substrate solution dispensers	2	MIL-S-51189
Alcohol tubes	20	B5-77-447
Benzene tubes	30	C5-77-444
Combination filters	75	C5-77-441

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3.1.2.1 Reagent container set. The reagent container set shall be as specified in table III. The ampules shall be "onion skin" tubing, (see 6.4). The ampules shall be sealed and shall be flat or concave on the sealed end and rounded on the outer end (see figure 1) and shall not leak when tested as specified in 4.3.4.4(a). Each ampule shall be wrapped one time with a 0.002 inch thick polyethylene sheeting conforming to L-P-378. Each wrap shall remain open on each end of the ampule and extend a minimum of 1/4 inch beyond the ends of the ampule. No more than 5 ampules shall be used in any one reagent container. All reagent containers which contain ampules shall have a plug of clean glass wool in the top. A sample of the glass wool treated with a few drops of H_2SO_4 shall show no discoloration when tested as specified in 4.3.4.3. The plug shall not be loosened by a rapid shaking motion. The tips of reagent containers (type A only) shall be wiped with a 2 percent ethereal solution of silicone fluid (see 6.9) to enable the container to dispense a symmetrical drop of liquid at the tip (see figure 1). All type A containers shall be sealed with a self-shrinking plastic seal applied so that it extends over the cap and container shoulder (see 6.6). The bromthymol blue ampule shall be a 10 ml prescored sealed ampule (see 6.11) containing 7 ml of bromthymol blue solution, MIL-B-11845, for reagent container 13. The prescored ampule shall not leak when tested as specified in 4.3.4.4(b).

3.1.2.2 Modified Dragendorff papers. Dragendorff papers shall consist of filter paper, as specified on Drawing B5-77-442, impregnated with the following solution:

The solution is prepared by mixing 15.2 g of metaphosphoric acid into 100 ml of distilled water, then adding and stirring until dissolved 6.7 g of cadmium iodide and 8.6 g of calcium chloride, and finally adding 20 g of bismuth oxychloride in small amounts while stirring. After the last addition, stirring shall be continued for 20 minutes. The black precipitate that forms shall be filtered off. The filtrate shall have a yellow or yellow-green color. (The filtrate shall be discarded if it has an orange or reddish-orange color.) To the filtrate 20 g of sorbitol shall be added and mixed until dissolved.

For each 20 ml of solution, 20 papers only shall be dipped (using teflon-tip forceps) and dried in warm air at approximately 70° F and shall be dipped within 20-30 minutes after the solution has been prepared. The papers shall have a yellow-green color and shall not discolor (e.g., bleach to white or become mottled) when aged for two weeks at 65° ± 2° C. (Papers with orange color or having black spots shall be discarded.) An orange-red color shall develop when tested as specified in 4.3.4.1(c). The papers shall then be assembled as specified on Drawing B5-77-442.

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TABLE III. Reagent container set

Reagent No.	req'd	Container type*	Contents	Specification	Quantity
1****	1	C	Compound 34 tablets	MIL-T-51023	14 tablets
2****	1	B	Sodium Pyrophosphate peroxide	MIL-P-50014	14 packets
3	1	A	None	---	---
4****	1	**	Substrate solution-dispenser	MIL-S-51189	---
5	1	A	Tris-(hydroxymethyl)-aminomethane (see 6.5)	0.05 Molar pH 8.0 + 0.01	8 ml
6	1	A	None	---	---
7	1	A	Modified Dragendorff reagent	3.1.1.1(a)	2.5 ml***
			Bismuth oxychloride	Reagent grade	0.09-0.1 g
8	1	A	Sodium hydroxide	O-C-265	0.05 g***
9	1	A	Sodium alizarin sulfonate	O-C-265	1.3-1.5mg***
			Thorium nitrate	O-C-265	3.5-3.7mg***
			Potassium acid phthalate	O-C-265	36 -38mg***
			Glacial acetic acid	O-C-265	1.4-1.5ml***
10	3	A	CC_2	MIL-I-285	0.04 g***
			Ethyl alcohol	O-C-265	2 ml***
11	3	A	Gamma picoline	MIL-C-51130	0.66 ml***
			1-Phenyl-3-methyl-5-pyrazolone	MIL-C-51130	0.44 g***
			Ethyl alcohol	O-C-265	1.4 ml***
12	2	A	Sodium hydroxide	O-C-265	0.40-0.43 g***
13	1	A	Bromthymol blue solution	MIL-B-11845	---
			None (see 3.4.1(c), 4.3.4.1(f), 6.12)		
14	3	A	Di(p-biphenyl) thiocarbazone	197-54-505	0.001 g***
			Acetone	O-C-265	2 ml***

* Container A: Dwg C5-77-456
Container B: Dwg B5-77-485
Container C: Dwg B5-77-448

** This item shall be marked with the number 4 on cap and side with fluorescent paint, color signal green enamel, (see 6.7).

*** All chemicals which are asterisked shall be placed in one or more onion skin ampules as required prior to insertion into the reagent containers, (see 6.4).

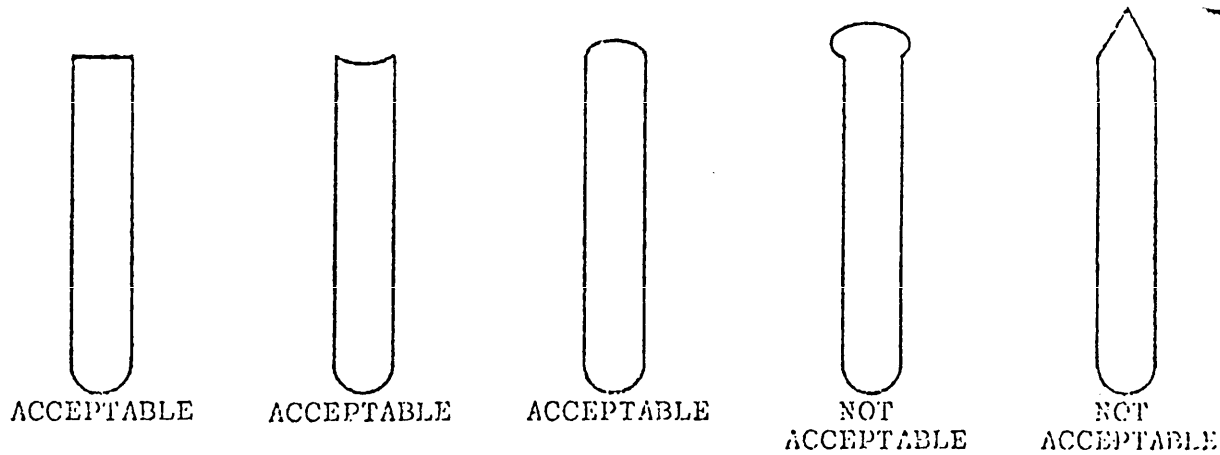
**** To be removed from the reagent container set, (see 3.4.1 and 3.4.6 and 6.10).

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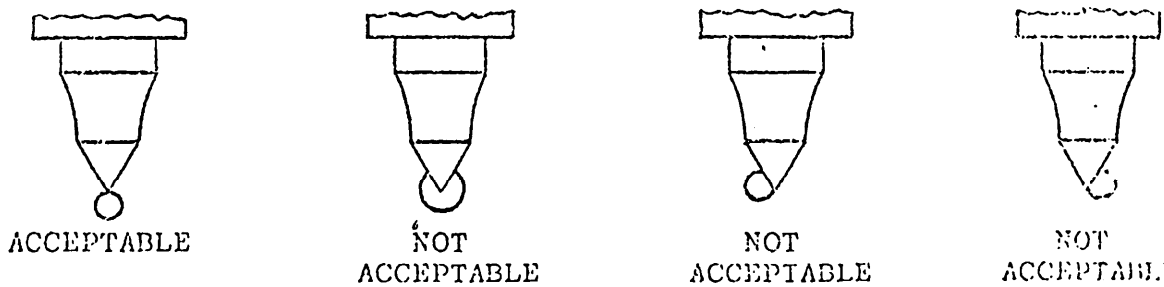
TABLE III. Reagent container set (Cont'd)

Reagent No.	req'd	Container type*	Contents	Specification	Quantity
15	2	A	Ferrous sulphate septahydrate	O-C-265	0.28 g***
			Iron wire	O-C-265	0.003 g***
16	1	A	Sulfuric acid, Sp Gr 1.38	O-C-265	1 ml***
17	1	A	Nitric acid, conc	O-C-265	2 ml***
18	3	A	DB3	MIL-D-50028	0.1 g***
			Acetone	O-C-265	2 ml***
19	1	A	Ammonium hydroxide, conc	O-C-265	2 ml***
20	2	A	Mercuric bromide	O-C-265	0.03 g***
			4,4'Bis(diethylamino) benzophenone	3.1.1.1(b)	0.01 g***
			Acetone	O-C-265	2 ml***
21	1	A	Hydroxylamine hydrochloride	O-C-265	0.5 g***
22	2	A	4-Nitronaphthalenediazo-amino 4'-azobenzene	Reagent grade	0.008 g***
			Ethyl alcohol	O-C-265	2 ml***
23	1	A	p, p' Tetramethyldiaminodi-phenylmethane	MIL-C-51130	0.1 g***
		A	Ethyl alcohol	O-C-265	2 ml***
24	1	A	Sodium hydroxide	O-C-265	0.5 g***
			Sodium sulfite	O-C-265	0.5 g***
25	1	A	Acetic acid, 1.0 N	O-C-265	1 ml***
26	2	A	Tetrabromophthaleinethylester	Reagent grade	0.002 g***
			Ethyl alcohol	O-C-265	2 ml***
27	1	A	None	---	---
28	2	B	Sodium perborate,tetrahydrate	3.1.1.1(c)	15 packets
29	1	A	Hydrochloric acid, conc	O-C-265	0.75 ml***
			Sodium molybdate, dihydrate	O-C-265	0.44 g***
			Ortho-dianisidine dihydrochloride	3.1.1.1(d)	0.008 g***
30	1	A	Neotetrazolium chloride	Reagent grade	0.02 g***
			Ethyl alcohol	O-C-265	2 ml***

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AMPULE SEALING



REAGENT CONTAINER DISPERSING TIP

Figure 1

AMPULE SEALING AND REAGENT CONTAINER DISPERSING CONFIGURATION

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3.2 Manufacture and assembly. All components shall be manufactured and assembled in accordance with their respective drawings.

3.2.1 Vapor leakage (alcohol tubes and benzene tubes). After assembly the alcohol tubes and benzene tubes shall show no loss in weig when tested as specified in 4.3.4.5.

3.2.2 Washing of combination filters. After assembly, the combinati filter shall be washed with a 50/50 (by volume) mixture of ethyl alcoho and xylene in a chromatography column. After washing, the combination filter shall be dried at room temperature in a dust-free atmosphere. Any handling of the combination filter after washing shall be by teflon tipped forceps only. The combination filter shall be considered satisfactory when:

(a) A gray-green or gray color develops on the glass fiber portion (inner circle) when tested with di(p-biphenyl) thiocarbazone (D in accordance with 4.3.4.2(a). A tan, black, red, pink or blue color that persists at the end of 90 seconds indicates contamination.

(b) A colorless or white to yellow color develops on the What man portion (outside ring) of the combination filter when tested with D in accordance with 4.3.4.2(b). A peach-brown or blue-purple color or specks indicates contamination.

3.3 Performance.

3.3.1 Reagent container set. The reagent container set (see table shall be aged at 65° ± 2° C for two weeks and shall pass the applicabl performance requirements in table IV when tested as specified herein.

3.3.1.1 Delivery volume. Reagent containers containing 4 or more a ules shall deliver a minimum of 65 percent of the original stated volu at 68° ± 2° F. All other reagent containers shall deliver a minimum o 75 percent.

3.3.2 Delivery volume of benzene tube. The benzene tube shall deli 1.5 ± 0.2 ml at 68° ± 2° F when tested as specified in 4.3.4.6.

3.4 Component assembly (table II).

3.4.1 Reagent containers, set. Each reagent container set, shall b individually packaged in a paperboard set-up box (0.040 inch nominal stock thickness) conforming to type I, grade A, class B of PPP-B-676.

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TABLE IV. Performance test requirements

Reagent No.	Reactant	Color or positive test	Blank	Test paragraph
7	$(CH_3)_4NBr$ or $(CH_3)_4NI$	Orange-red ppt or Solution	Clear and yellow Yellow-orange or yellow-green w/wo white ppt.	4.3.4.1(b)
8, 9	GB	Yellow	Pink	4.3.4.1(d)
10, 11	GA	Red-purple	Colorless	4.3.4.1(e)
13	pH4 pH7 pH10	Yellow Green Blue	--- --- ---	4.3.4.1(f)
14	Zn^{+}+ ions	Red-purple	Gray-green or blue-green	4.3.4.1(g)
12, 15, 16	GA	Blue	Colorless	4.3.4.1(h)
12, 17	DM	Green; yellow; red or dark pink*	Colorless	4.3.4.1(i)
18, 19	CN	Blue, purple-blue, or purple	Colorless	4.3.4.1(j)
20	HD	Red or red-brown	Colorless or pale yellow	4.3.4.1(k)
21	SeO_2	Red-brown, orange-brown, orange-red, brown	Colorless	4.3.4.1(l)
12, 22	CA^{++}ions	Pink, red, or red-purple	Blue or purple-blue	4.3.4.1(m)
23	Chloranil	Blue	Colorless	4.3.4.1(n)
24	PS	Red-purple	Colorless	4.3.4.1(o)
25, 26	Bovine Albumine Serum, water soluble	Blue-green	Yellow	4.3.4.1(p)
28, 29	GA	Red-brown or tan-brown w/ or w/o ppt	Colorless, light blue or yellow w/ or w/o ppt	4.3.4.1(q)
8, 30	Benzenethiol	Red or purple-brown	Colorless or yellow-brown	4.3.4.1(r)

*High concentration give a green color after addition of 17; yellow color after addition of 12. Final color must be red or dark pink.

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The box shall have inside dimensions of approximately 8 inches by 4-3/4 inches by 2 inches. The box shall be divided into 42 equal cells extending the full depth of the box by means of interlocking partitions made from the same material as the box.

(a) Vials 1 and 2 shall be removed from each reagent container set and then both vials of each type shall be individually wrapped then packaged together, method IA-8 of MIL-P-116.

(b) Each cell of each reagent set from which vials have been removed shall be filled by a plug of either polyurethane foam, 1 to 2 PCF, type I, class 2 of MIL-P-26514 or foamed rubber, MIL-R-5001 or MIL-R-20092, type II medium.

(c) The contents for the empty reagent No. 13 bottle, that is the bromthymol blue solution, shall be contained in a prescored 10 ml ampule (see 6.11). Two of these ampules (1 for each reagent container set in the kit) shall be sealed and packaged together and stored separately from the empty reagent No. 13 bottle in the reagent container sets. The plastic squeeze bottle, reagent No. 13 container in the reagent container set, shall remain empty until prior to use (see 4.3.4.1(f)) Each prescored ampule shall be completely enclosed in cellulosic cushioning material conforming to type I, class B of PPP-C-843 and secured with tape conforming to PPP-T-60. The two individually wrapped ampules shall be placed in a 3-1/4 by 2-11/16 by 1-3/16 inches, 0.020 inch nominal thickness paperboard box conforming to variety 2, style II, type A, class a of PPP-B-566. Additional cushioning material shall be included as necessary, to restrict free movement of the cushioned ampules. The box shall be enclosed in heat sealed barrier bag conforming to type II, class E, style 1 of MIL-B-117.

3.4.2 Modified Dragendorff papers. Seventy-five modified Dragendorff paper assemblies, conforming to Drawing B5-77-442, shall be packaged in a paperboard folding box (0.026 inch nominal stock thickness) conforming to variety 2, style IV, type A, class a of PPP-B-566. The box shall have inside dimensions of approximately 2 inches by 2 inches by 2 inches.

3.4.3 Alcohol tubes. Twenty alcohol tubes, each conforming to Edgewood Arsenal Drawing B5-77-447 shall be individually enclosed in a single, type III, style I of PPP-P-291. The tube shall be wrapped with the flutes of the paperboard aligned parallel to the length of the tube. The dimension of the paperboard measured paralled to the flutes shall be 5-1/8 $\pm$ 1/8 inch and its dimension measured at right angles to the flutes shall be 5-1/4 $\pm$ 1/8 inch. Self-adhering material meeting the requirements of

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the corrugated paperboard specified herein, may be used to eliminate the use of tape; otherwise the wrap of corrugated paperboard shall be secured around the tube by tape conforming to class 2 or 3 of PPP-T-60. Secure ends of wrap with tape to prevent spillage of contents.

3.4.4 Benzene tubes. Thirty benzene tubes, conforming to Drawing C5-77-444, part number C5-77-444-1, shall be packaged blind end down in a paperboard box (0.033 inch nominal stock thickness) conforming to variety 2, type I, grade A, class B of PPP-B-676. The box shall have inside dimensions of approximately 2-1/2 by 3 by 3-1/4 inches with partitions 3 inches high. The partitions shall form 30 equal wells by means of interlocking slotted partitions made from the same material as the box. Three benzene tube adapter assemblies conforming to Drawing C5-77-444, part number C5-77-444-2, shall be placed in any three of the wells along with the benzene tubes. A direction card conforming to Drawing C5-77-444, part number C5-77-444-3, shall be placed flat on the top of the partitions, printed side up. Cushioning material conforming to polyurethane foam 1/4 inch thick, 1 or 2 PCF, type I, class 2, of MIL-P-26514 shall be included in the bottom of the box.

3.4.5 Combination filters. Seventy-five combination filter assemblies, conforming to Drawing C5-77-441, shall be packaged in a box as specified in 3.4.2.

3.4.6 Substrate dispensers. Two substrate dispensers, reagent 4, shall be packaged in a snug fitting box conforming to type II, grade A, class C, style 4 of PPP-B-676.

3.5 Workmanship. The kit components shall be free from damage (cracked or broken ampules or containers, punctured tubes, ripped or torn detector papers or filters) and contamination (grease, oil or foreign matter).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1.1, 3.1.1.1(b), 3.1.1.1(c), 3.1.1.2, 3.1.2, 3.1.2.1, 3.1.2.2, 3.2.2, 3.3.1.1, 3.4 and section 5 of this specification have been satisfied.

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of individual components produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of basic chemicals or materials shall be represented in any lot of finished components.

4.2.2 Sampling.

4.2.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For destructive tests. Sampling shall be conducted in accordance with MIL-STD-105, level S-4. Six reagent bottles shall be selected from each lot for the glass wool plug test in accordance with 4.3.4.3.

4.3.3 Inspection procedure.

4.3.3.1 For examination and nondestructive tests. Sample items shall be examined in accordance with the classification of defects and MIL-STD-105.

4.3.3.2 For destructive tests. Sample items shall be tested in accordance with 4.3.4. The acceptance number is zero.

4.3.3.3 Classification of defects.

(a) Reagents, solvents, and test papers for sampling and analyzing kit, CBR agents.

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Component missing
102	Marking illegible
103	Marking
104	Marking missing
105	Workmanship (3.5)

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4.3.4 Tests. Distilled water from the polyethylene bottles furnished with the kit shall be added to the reagent bottles in accordance with table V. Reagent grade chemicals shall be used throughout the test. Blank determinations shall be run and corrections applied where significant. The blank shall consist of all the reagent except the reactant.

TABLE V. Distilled water addition

Reagent No.	Distilled water ml
8	5
9	3.5-3.6
12	5
15	3
16	4
21	5
23	2
24	5
25	4
29	4.5

The following procedure shall apply in all cases where glass fiber or Whatman paper is specified in the test: The Whatman or glass fiber portion of the combination filter shall first be cut into four equal sections. The test shall then be run on one of the sections placed in the micro test tubes. All handling of the test papers shall be done with the plastic tipped forceps or plastic needle. By means of the plastic needle, lift the test paper close to the mouth of the test tube needle to push the test section down into the micro test tube for heating. When heating is specified, the micro test tubes shall be held in the adapter from the alcohol lamp and heated in a laboratory steam bath. Where no heat is required, the test paper may be held with the plastic tipped forceps. Test shall be conducted as follows:

4.3.4.1 Reagents.

4.3.4.1(a) ODN.HCl purity and transmittance.

4.3.4.1(a)(1) Purity. Weigh out four specimens (15 to 45 mg) of Government furnished ODN.HCl to the nearest 0.1 mg and dissolve each specimen in 20 ml of water. Adjust the pH to less than 3 with hydrochloric acid and cool below 10° C. Diazotize by adding a one-percent sodium nitrite solution until potassium iodide starch-paper turns blue;

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dilute to volume in a 100 ml volumetric flask. Take a suitable aliquot (2 ml) and place it in another 100 ml volumetric flask. Cool the aliquot below 10° C and add a two-percent solution of resorcinol in ten percent sodium hydroxide until color formation is complete. Dilute the volume and read percent transmittance in a spectrophotometer at 510 millimicrons. Run a blank using one-percent sodium nitrite. Plot a graph on semi-log paper, of concentration (milligrams ODN.HCl) versus percent transmittance. To check the unknown specimens weigh out the unknown and treat as above. From the standard graph obtained above, determine the amount of ODN.HCl present and calculate the percent purity as follows:

$$\text{Percent purity} = \frac{100A}{W}$$

where: A = Amount determined from standard graph.
W = Weight of unknown specimen.

4.3.4.1(a)(2) Transmittance. Dissolve 8.00 ± 0.50 mg of ODN.HCl and 2.000 ± 0.200 g of sodium pyrophosphate peroxide in 100 ml of water. Filter the solution and age for 24 hours at 38° ± 1° C. Read percent transmittance in a spectrophotometer at 510 millimicrons.

4.3.4.1(b) Reagent 7 (Modified Dragendorff Reagent). Add one drop of a one-percent solution of either tetramethyl ammonium iodide or tetramethyl ammonium bromide in water to 0.1 ml of reagent 7 in a micro test tube. Record the color developed. Note: Prior to performing the test, the reagent 7 ampule should be crushed, shaken for a few minutes and allowed to stand for 50 minutes prior to use.

4.3.4.1(c) Modified Dragendorff Paper. Deposit one drop of a one-percent solution of either tetramethyl ammonium iodide or tetramethyl ammonium bromide in water on a Dragendorff paper; record the color developed. Observe for discoloration, bleaching or mottled appearance of the aged paper before performing test.

4.3.4.1(d) Reagents 8 and 9. Prepare a test sample of 0.1 ml of aqueous solution containing 50 micrograms (ug) of GB. Add two drops of reagent 8 to the test sample. Allow the test tube to stand approximately one minute. Add three drops of reagent 9 to the test tube and agitate. Record the color developed. Note: Prior to performing the test, allow 15 minutes for reagent 9 to dissolve. The first few drops should be discarded before use. Reagent 9 should be pink.

4.3.4.1(e) Reagents 10 and 11. Prepare a test sample of 0.1 ml of a freshly prepared aqueous GA solution (containing 0.5 to 1.5 ug of GA).

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Add one drop of reagent 10, shake, and one drop of reagent 11 to the sample in a test tube. Record the color developed.

4.3.4.1(f) Reagent 13. Prepare buffered solutions of pH4 (acidic), pH7 (neutral), and pH10 (basic). To one drop of each buffered solution in a separate test tube, add one drop of reagent 13 and observe the color developed in each tube. Note: Prior to performing this test the prescored ampule containing the bromthymol blue solution shall be broken at the score line and the contents transferred to the number 13 reagent squeeze bottle of the reagent container set.

4.3.4.1(g) Reagent 14. Dissolve sufficient zinc salt (such as zinc sulfate) in 100 ml of water so that the solution contains 0.03 g of zinc ions. Deposit one drop (0.05 ml) of the zinc ion solution on the glass fiber paper portion of the combination filter and allow water to evaporate. Add one drop of reagent 14 to the glass fiber paper, allow the solvent to evaporate and observe the color developed after 90 seconds from the time of the addition of reagent 14.

4.3.4.1(h) Reagents 12, 15 and 16. Dissolve 0.07 g of GA in 100 ml of benzene. Deposit one drop (0.05 ml) of GA solution on Whatman test paper in a micro test tube. Add one drop of reagent 12 to the test section and heat for one minute in the steam bath. Add one drop of reagent 15 to the test section. After one minute, add one drop of reagent 16 to the test section. Heat for 2 minutes in the steam bath. Repeat the addition of reagent 16 and subsequent heating until the test section is colorless or blue in color. Do not repeat more than twice.

4.3.4.1(i) Reagents 12 and 17. Prepare a test solution by dissolving 0.02 g of DM in 100 ml of benzene. Deposit one drop (0.05 ml) of the DM solution on Whatman paper in a micro test tube and allow the benzene to evaporate. Add one drop of reagent 17 to the Whatman paper and heat for 10 minutes in a steam bath and add one drop of reagent 12. Observe the color developed. If color does not appear repeat the one drop addition of reagent 12 up to four times.

4.3.4.1(j) Reagent 18 and 19. Prepare a test solution by dissolving 0.016 g of CN in 100 ml of benzene. Deposit one drop (0.05 ml) of CN solution on Whatman paper in a micro test tube and allow the solvent to evaporate. Add one drop of reagent 18 to the test paper. Add one drop of reagent 19 to the test paper and heat for 3 minutes in a steam bath. Add one additional drop of reagent 19 to the test paper and record the color developed.

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4.3.4.1(k) Reagent 20. Prepare a test solution by dissolving 0.02 g of HD in 100 ml of benzene. Deposit 1 drop (0.05 ml) of the HD solution in a micro test tube containing a Whatman and glass fiber paper. Add one drop of reagent 20 to the test paper. Heat for 2 minutes in a steam bath and record the color developed.

4.3.4.1(l) Reagent 21. Prepare a test solution by dissolving 0.024 g of selenium dioxide in 100 ml of water. Deposit one drop of the selenium dioxide solution on glass fiber paper in a micro test tube and allow the water to evaporate. Add one drop of reagent 21 to the test paper. Heat to dryness on a steam bath and record the color developed.

4.3.4.1(m) Reagents 12 and 22. Prepare a test solution by dissolving sufficient cadmium chloride in 100 ml of water so that the solution contains 0.03 g of cadmium ions. Deposit one drop (0.05 ml) of the cadmium ion solution on glass fiber paper in a micro test tube and allow the solvent to evaporate. Add one drop of reagent 22 to the test solution and allow the solvent to evaporate. Add one drop reagent 12 to the paper and record the color developed. Note: Allow sufficient time for reagent 22 to dissolve prior to use.

4.3.4.1(n) Reagent 23. Add one drop of reagent 23 to one drop (0.05 ml) of chloranil solution on a porcelain spot plate. Record the color developed. Note: Chloranil solution is prepared by dissolving 0.01 g chloranil (see 6.8) in 100 ml of 1:1 ethyl alcohol. This solution should be freshly prepared and used within 4 hours.

4.3.4.1(o) Reagent 24. Using a Q5 toxic gas dilution apparatus (Drawing DLB136-41-780-1) expose a type I detector tube to 10 micrograms of PS using a concentration of 10-15 microgram per liter at a 1 liter per minute flow rate. Add two drops of reagent 24 to the gel in the detector tube. Observe the color of the gel after 15 seconds.

4.3.4.1(p) Reagents 25 and 26. Prepare a test solution by dissolving 0.06 g of water soluble bovine albumine serum in 100 ml of water. Deposit one drop of (0.05 ml) of the bovine albumine serum solution on glass fiber paper in a micro test tube and allow the water to evaporate. Add one drop of reagent 26 to the test paper and then add three drops of reagent 25 to the test paper and record the color developed.

4.3.4.1(q) Reagents 28 and 29. Prepare a test solution by dissolving 0.06 g of GA in 100 ml of ethyl alcohol. Empty the contents of one packet of reagent 28 into a clean, dry micro test tube. Add one drop (0.05 ml) of the GA solution to the test tube so that the drop is deposited on

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the reagent 28 powder (without running down the sidewall of the test tube). Wash down the sidewall of the test tube with one to two drops of ethyl alcohol. Evaporate the contents of the test tube to dryness by means of a hot water bath. Remove the test tube from the water bath, heat the contents of the test tube over the flame of an alcohol burner for 3-5 minutes and allow the test tube to air-cool. When the test tube is cool to the touch, add two drops of reagent 29. Record the color developed.

4.3.4.1(r) Reagents 8 and 30. Prepare a test solution by dissolving 0.03 g of benzenethiol in 100 ml of ethyl alcohol. Place one drop (0.05 ml) of the benzenethiol solution in a micro test tube. Add two drops of reagent 8 to the test tube and wait one minute. Add two drops of reagent 30, wait four minutes and record the color developed. If the color is yellow or yellow brown, add one more drop of reagent 30. Record the color developed.

4.3.4.2 Combination filter; after washing.

(a) Add one drop of reagent 14 to the glass fiber portion of the combination filter, and observe the color at the end of 90 seconds.

(b) Add one drop of reagent 18 and one drop of concentrated ammonium hydroxide on the Whatman portion of the combination filter. Heat the filter paper for three minutes in a hot water bath. Add one additional drop of concentrated ammonium hydroxide. Observe the color developed.

4.3.4.3 Glass wool plug. Remove the plastic tip from the reagent bottle and add three drops of concentrated, reagent grade, H_2SO_4 to the glass wool plug in the tip. Observe the glass wool for any discoloration.

4.3.4.4 Leakage.

4.3.4.4(a) Onion skin ampule. Place the ampules in a vacuum desiccator and apply a vacuum of 5 inches of mercury. Close the stopcock and maintain the vacuum for 30 minutes. At the end of this period, draw in sufficient colored water (0.1g 3,3' - di-bromothymolsulfonphthalein and 8.0 ml N/50 sodium hydroxide per 2.5 liters) to cover the ampules. Release the vacuum. Wait 5 minutes and examine ampules for evidence of water penetration.

4.3.4.4(b) Prescored ampule. The ampule shall be placed under water at 25° ± 3° F above ambient temperature for 3 minutes, with the sealed end facing upward. An emission of bubbles shall be considered as leakage.

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4.3.4.5 Vapor leakage. Weigh the alcohol tube or benzene tube to the nearest milligram and place in a constant temperature oven at 65° ± 2° C for 24 hours. Reweigh to the nearest milligram. Any loss in weight shall be considered vapor leakage.

4.3.4.6 Delivery volume of benzene tube. Determine the delivery volume of the benzene tube by squeezing the contents of the tube into a micro-buret and reading the scale.

5. PREPARATION FOR DELIVERY

5.1 Packaging, commercial. The components for the analyzing kit shall be packaged in accordance with FED-STD-356.

5.2 Packing, commercial. Components for the analyzing kit, packaged as specified in 5.1, shall be packed in accordance with FED-STD-356.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with FED-STD-356.

6. NOTES

6.1 Intended use. This specification covers chemical reagents, solvents, and test papers used in the Sampling and Analyzing Kit, CBR Agents, M19.

6.2 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.3 Ortho-dianisidine dihydrochloride (ODN.HCl). o-Dianisidine dihydrochloride is prepared by dissolving 15 g technical grade o-dianisidine (melting point 137° to 140° C) in 150 ml of water, adding 11.25 ml of concentrated hydrochloric acid, and heating the solution to boiling. The next step is to add 0.6 g of activated charcoal to the solution and boil for 8 minutes; then 0.75 ml of 32 percent aqueous sodium bisulfite solution is added and the solution is boiled for an additional 2 minutes and filtered through a Buchner funnel using acid-resistant filter paper. The additions of charcoal and sodium bisulfite, the boiling, and the filtration procedure is repeated three times. To the filtrate, 30 ml of concentrated hydrochloric acid containing 0.75 ml of 32 percent aqueous sodium bisulfite solution is added. The filtrate is then cooled to approximately room temperature, and the precipitate is filtered off. (Note: No more than 30 minutes shall elapse

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between the start of cooling the filtrate and the collecting of the precipitate.) The collected precipitate is washed with 80 percent ethanol and then with absolute alcohol. The collected precipitate is vacuum-dried at 50° C to obtain the white product.

6.4 Tubing. The "onion skin" tubing used in making the ampules is designated as Kimble Glass Co. No. R-6 glass, with an outside diameter of 5 to 7 millimeters. Kimble Glass Co., Vineland, New Jersey.

6.5 Tris-(hydroxymethyl)-aminomethane. Tris-(hydroxymethyl)-aminomethane solution is prepared by dissolving 0.607 g of tris (hydroxymethyl) aminomethane in 25 ml of water, adding 28.7 ml of 0.1N hydrochloric acid, and diluting to 100 ml. The pH is adjusted to 8.0.

6.6 Plastic seal. DuPont "Cel-O-Seal" band, 20/25 style TRST, Plain, has been found to be satisfactory. After application, the reagent container is dried in an oven for 25 to 30 minutes at 60° to 65° C.

6.7 Fluorescent paint. Fluorescent paint, from Day-Glo Color Corp., Cleveland, Ohio 44103, has been found to be satisfactory.

6.8 Chloranil. Eastman Organic Chemical Catalogue, Chloranil P603.

6.9 Silicone fluid. Dow-Corning 200 fluid (viscosity 200 cps at 25°C).

6.10 Reagent container set (table III). Due to the incompatability of items, Ident. No. 1, 2, and 4, with other items in the set, they will be assembled separately (see 3.4.1 and 3.4.6).

6.11 Prescored ampule. Kimble Glass Co., 10 ml prescored sealed ampule has been found to be satisfactory.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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